Theory

ISPhO

A1-1

English

Ferromagnetic domain wall dynamics

Part A. Single magnetic moment (2.0 points)

A.1 (0.3pt)

A.2 (0.9pt)

$m_z(t) =$

A.3 (0.8pt)

$\Omega =$

Part B. Static domain wall (3.5 points)

B.1 (0.5pt)

(Formula) $B_{\mathrm{ex}} =$

(Value) $B_{\mathrm{ex}} =$

(Formula) $B_{\mathrm{an}} =$

(Value) $B_{\mathrm{an}} =$

(Formula) $\omega =$

(Value) $\omega =$

B.2 (1.0pt)

$\alpha =$

$T =$

$d =$

B.3 (1.5pt)

β	d

B.4 (0.5pt)

$c =$

Part C. Domain wall motion in external field (4.5 points)

C.1 (0.2pt)

Direction (to the right / to the left):

C.2 (2.0pt)

b	v	φ

C.3 (0.4pt)
Plot:

$dv/db =$

C.4 (0.6pt)
Plot:

$v_c =$ $b_{\mathrm{cr}} =$

C.5 (0.3pt)

$v_c =$ $B_{\mathrm{cr}} =$

C.6 (1.0pt)

t	x

Plot:

Theory

ISPhO

A2-1

English

Vibrational structure of polyynes

Part A. An infinite chain of carbon atoms (6.5 points)

A.1 (0.8pt)

$E =$

A.2 (0.8pt)

$m\ddot{u}_n =$

$m\ddot{v}_n =$

A.3 (0.9pt)

$a =$ $b =$

$c =$ $d =$

A.4 (1.0pt)

$\omega_1/\omega_0 =$

$\omega_2/\omega_0 =$

A.5 (1.0pt)

$\omega_1/\omega_0 \approx$

$\omega_2/\omega_0 \approx$

A.6 (1.0pt)

$\omega_1/\omega_0 \approx$

$\omega_2/\omega_0 \approx$

A.7 (1.0pt)
Plot:

$\alpha = 0.5,\ \beta = 0.1$	$\alpha = 0.5,\ \beta = 3.0$

Part B. Spectroscopy of polyynes (3.5 points)

B.1 (0.8pt)

Equation:

B.2 (0.7pt)

$q_l =$

B.3 (2.0pt)

Single bond:

Triple bond:

Single-electron transistor

Part A. Tunneling effect (3.0 points)

A.1 (0.1pt)

$W =$ $\Delta W =$

A.2 (0.3pt)

$q_g =$ $q_j =$

A.3 (0.3pt)

$\Delta E =$

A.4 (0.5pt)

$A =$ $Q =$

A.5 (0.5pt)

$I_t =$

Condition on n:

A.6 (0.3pt)

$I =$

A.7 (1.0pt)
Plot:

Part B. Single-electron transistor (4.5 points)

B.1 (0.2pt)

$\varphi =$

B.2 (1.0pt)

$A =$

B.3 (1.0pt)

$Q =$

B.4 (0.3pt)

Condition:

B.5 (1.5pt)

Conditions:

Plot:

B.6 (0.5pt)

$I(V_g, V) =$

Part C. Tunneling at finite temperature (2.5 points)

C.1 (0.5pt)

$C_1 = C_2 =$ $C_g =$

C.2 (0.3pt)

$dp(n)/dt =$

C.3 (0.5pt)

$p(n+1) =$

C.4 (0.3pt)

C.5 (0.5pt)

$p(0) =$ $p(1) =$

C.6 (0.4pt)

$I =$